MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, HIGH-FREQUENCY

DEVICE TYPES: JAN 2N6603 AND TXV 2N6603
JAN 2N6604 AND TXV 2N6604

This specification is approved for use by the Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN, Silicon, Microwave transistors. The prefix "JAN" is used on devices submitted to and passing the Precap Visual and Bond Strength testing as specified in 4.5. The prefix "TXV" is used on devices submitted to and passing the inspection specified in 4.5 and 4.6.

1.2 Physical Dimensions. See Figure 1

1.3 Maximum Ratings.

	P_T				V_{CBO}	V_{CEO}	V_{EBO}	I_C	T_j	T_{stg}
	1/ T_A=25°C	3/ T_C=100°C	2/ T_A=25°C	4/ T_C=75°C						
	mW	mW	mW	mW	Vdc	Vdc	Vdc	mAdc	°C	°C
2N6603	400	400	---	---	25	15	3.0	30	200	-65 to + 200
2N6604	---	---	500	500	25	15	3.0	50	200	-65 to + 200

1/ Derate linearly 2.3 mW/°C for T_A > 25°C
2/ Derate linearly 2.9 mW/°C for T_A > 25°C
3/ Derate linearly 4 mW/°C for T_C > 100°C
4/ Derate linearly 4 mW/°C for T_C > 75°C

1.4 Primary Electrical Characteristics.

	h_{FE}				C_{cb}		NF		G_{pe}			
	V_{CE}=10Vdc I_C=15mAdc		V_{CE}=10Vdc I_C=30mAdc		V_{CB}=10Vdc I_E=0 .1MHz≤f≤1MHz		V_{CE}=10Vdc I_C=5mAdc f=1GHz		V_{CE}=10Vdc I_C=15mAdc f=1GHz		V_{CE}=10Vdc I_C=30mAdc f=1GHz	
	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX
	---	---	---	---	PF	PF	dB	dB	dB	dB	dB	dB
2N6603	30	200	---	---	.25	.75	1.0	2.5	14	20	---	---
2N6604	---	---	30	200	.45	.95	1.5	3.0	---	---	14	20

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:44Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts
MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, Symbols, and Definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500, and as follows:

G_{NF} - - - - - - Power gain with device tuned for optimum noise figure.

3.3 Design, Construction, and Physical Dimensions. Transistors shall be of the design, construction, and physical dimensions shown on Figure 1.The individual device shall be supplied in a suitabe device carrier.

3.4 Performance Characteristics. Performance characteristics shall be as specified in Tables I, II and III, and as follows:

3.4.1 Process-Conditioning, Testing, and Screening For "TXV" Types. Process-conditioning, testing, and screening for the "TXV" types shall be as specified in 4.5 and 4.6.

3.4.2 Internal Visual (PRECAP) Inspection. All devices shall in addition to all performance requirements, be internally visually inspected in accordance with 4.5.1.

3.5 Marking. All device marking specified in MIL-S-19500 may be omitted from the body of the transistor except the applicable prefix and device type number however,all marking specified in MIL-S-19500 shall be marked on the device carrier. Examples of acceptable device marking are as follows:

J2N	JV2N
6603	6603

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:44Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and Inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification Inspection. Qualification inspection shall consist of the examinations and tests specified in Tables I, II, and III.

4.2.1 Qualification Testing. Any level, JAN or JANTXV may be used for qualification testing; however, the test report shall contain only that data specified in 4.2. Upon request to the qualifying activity, qualification will be extend to include the other device levels.

4.2.2 Variables Data Criteria For Qualification Inspection. For qualification inspection, Group A variables data shall be submitted on a sample associated with the lowest LTPD specified in Table I. However, in subgroups with higher LTPD's, acceptance criteria shall be based on a pre-designated portion of that sample. In addition, variables data shall also be submitted on the Group B and C pre-test and post-test endpoints as specified in Table II. The remaining Group B and C samples need not be submitted to all tests of Group A.

4.2.3 Thermal Shock (Temperature Cycling). Samples that have been subjected to the Group B thermal shock test may be continued on test to 25 cycles in order to satisfy Group C thermal Shock test requirements. These samples shall be predesignated, and shall remain subjected to the Group C 25 cycle acceptance evaluation after they have passed the Group B 10 cycle acceptance criteria. The cumulative total of failures found during the 10 cycle test and during the subsequent interval up to 25 cycles shall be computed for 25 cycle acceptance criteria. At the option of the manufacturer, Group B samples may be continued on test, without interruption for endpoint measurements, through Group C test requirements of 25 cycles.

4.3 Quality Conformance Inspection. Quality conformance inspection shall consist of Group A, B, and C inspections. When specified in the contract or order, one copy of the quality conformance inspection data, pertinent to the device inspection lot, shall be supplied with each shipment by the device manufacturer (see 6.2).

4.3.1 Group A Inspection. Group A inspection shall consist of the examinations and tests specified in Table 1.

4.3.2 Group B Inspection. Group B inspection shall consist of the examinations and tests specified in Table II.

4.3.3 Group C Inspection. Group C inspection shall consist of the examinations and tests specified in Table III. This inspection shall be conducted on the initial lot and thereafter every 12 months during production.

4.3.4 Group B and Group C Life-Test Samples. Samples that have been subjected to Group B, 340-hours life-test, may be continued on test to 1,000 hours in order to satisfy Group C life-test requirements. These samples shall be predesignated, and shall remain subjected to the Group C 1,000 hour acceptance evaluation after they have passed the Group B 340-hour acceptance criteria. The cumulative total of failures found during 340-hour test and during the subsequent interval up to 1,000 hours shall be computed for 1,000-hour acceptance criteria (see 4.3.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:44Z
Check the source to verify that this is the current version before use.

4.4 Methods Of Examination and Test. Methods of examination and test shall be as specified in Tables I and II, and as follows:

4.4.1 Pulse Measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.4.2 Power Gain, Noise Figure and G_{NF}. The device shall be tested in the circuits of figures 5 or 6. The specified conditions shall be applied and the slide screw tuners adjusted to obtain maximum power output for the G_{pe} test and minimum noise figure for NF and G_{NF} tests.

4.4.3 Resistance To Solvents. Transistors shall be subjected to tests in accordance with method 215 of MIL-STD-202. The following details shall apply:

(a) All areas of the transistor body where marking has been applied shall be brushed.

(b) After subjection to the tests there shall be no evidence of mechanical damage to the device and markings shall have remained legible.

4.5 Process-Conditioning, Testing, and Screening For "JAN" and "TXV" Types. Precap Visual and Bond Strength testing for "JAN" types shall be in accordance with 4.5.1 through 4.5.2.5. Process-conditioning, testing and screening for "TXV" types shall be in accordance with 4.5.1 through 4.6.9.1.

4.5.1 Internal Visual (PRECAP) Inspection and Process-Conditioning, Testing, and Screening. The internal visual inspection shall be performed in accordance with test method 2072 of MIL-STD-750 prior to encapsulation on a 100 percent basis. "TXV" devices shall also include process-conditioning, testing and screening as specified in 4.6. The manufacturer shall permit the authorized government representative to witness concurrent with time of manufacturer's performance of these tests, the process-conditioning, testing and screening of the devices. These conditioning and screening tests normally performed by a manufacturer as standard production tests, need not be repeated when these are predesignated and acceptable to the government as being equal to or more severe than the test specified herein.

4.5.2 Bond Strength Testing. Bond strength testing is performed to destructively measure and determine compliance with bond strength requirements. Those bond strength tests, normally performed by a manufacturer as standard production tests, need not be repeated when these are predesignated and acceptable to the Government as being equal to or more severe than the test specified herein.

4.5.2.1 Apparatus. The apparatus for this test shall consist of suitable equipment for applying the specified stress to the bonded wire loop of internal lead wires. A calibrated measurement and indication of the applied stress in grams-force (gf) at the point of failure shall be provided by equipment capable of measuring stresses within $\pm$2.5 percent of the indicated value.

4.5.2.2 Sample Inspection. A minimum of 8 wire bond loops from each lead bond machine (or pair of machines if two different ones are used in making the bonds on a single device) shall be pulled to destruction every two hours of operation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:44Z
Check the source to verify that this is the current version before use.

4.5.2.2 Sample Inspection Continued. A sample inspection of 8 bonded loops shall be performed at the beginning of each shift or operator changeover and after a product changeover, wire spool change, or bonding machine correction prior to release for production runs. A method of segregating, identifying, and securing units produced between inspections shall be employed.

4.5.2.3 Procedure. The device shall be secured so that the wires to be pulled are in an upward position. The pull test is performed by inserting a rounded hook under the span of wire midway between the die bonds and post bond then pulling vertically to destruction. At least half of the wires on the device shall be tested in this manner including at least one base and one emitter wire. If there is an odd number of wires then an extra wire shall be pulled. The force required to cause failure and the failure category shall be recorded. Any bond pulled which fails under an applied stress less than that indicated in Table IV or Figure 2 as the required minimum bond strength for the indicated wire diameter shall constitute a failure. Table IV gives the minimum acceptable breaking force values for the most common wire sizes. The minimum bond pull limits for other wire sizes may be obtained from the curve in Figure 2.

TABLE IV.

Wire Diameter (MILS)	Manufacturer's Minimum Bond Pull Limit (Grams Force).	
	Aluminum	Gold
.7	.7	.7
1.0	1.0	3.0
1.5	2.0	4.0
2.0	3.0	5.0

4.5.2.4 Failures Due To Die Fracture. If silicon from the semiconductor die is pulled off with the bond or die soldered joint, and provided the failure occurs at or below the minimum pull limits as specified in Table IV or Figure 2, then the entire lot produced on that equipment since the last acceptable sample was taken will be rejected. Since die fracture may indicate bonding damage to the silicon, a nondestructive pull test (4.5.2.5) cannot be used to qualify product failing in this mode.

4.5.2.5 Corrective Action and Product Disposition. If a sample contains one or more defects as described in 4.5.2.3, the machine (or machines, if two were used in making each bond pair) from which the sample was taken will not be allowed to produce additional "JAN" or "TXV" product for shipment until a sample has passed. All units wire bonded on the machine (or machines) that produced the defective(s) since the last acceptable sample inspection, will either be rejected or subject to a 100 percent nondestructive bond pull at a force of one-half the minimum bond pull limit (4.5.2.3) for the particular size wire. A nondestructive pull force value of $\frac{\bar{X} - 3\sigma}{2}$ may be substituted for this value, providing $\sigma \leq 0.2\bar{X}$.

Source: https://assist.dla.mil -- Downloaded: 202610-19T02:44Z
Check the source to verify that this is the currentversion before use.

4.5.2.5 Corrective Action and Product Disposition Continued. The statistical data for this will be obtained from actual pull data from the last full acceptable day's production before the defect was found. $\bar{X}$ is the average pull strength and σ is the standard deviation. If 99.999 percent pure aluminum annealed wire is used, the divisor in the equation shall be changed to 3 and the second sentence above will then read "1/3 the minimum bond pull limit".

4.6 Process-Conditioning, Testing, and Screening for "TXV" Types. The procedure for process-conditioning, testing, and screening for "TXV" types shall be in accordance with MIL-S-19500 and 4.6.1 through 4.6.9.1 and Figure 3. At the option of the manufacturer, "TXV" devices may be processed in accordance with Figure 4 and 4.6.1 thru 4.7. Those conditioning and screening tests normally performed by a manufacturer as standard production tests, need not be repeated when these are predesignated and acceptable to the government as being equal to or more severe than the test specified herein. (At the option of the manufacturer, the non-TXV type may be subjected to the process-conditioning and testing).

4.6.1 Quality Assurance (Lot Verification). Quality assurance shall be in accordance with MIL-S-19500 except lot records shall be kept for 2 years, minimum.

4.6.2 High-Temperature Storage. All devices shall be stored for at least 24 hours at a minimum temperature (T_A) of 200°C.

4.6.3 Thermal Shock (Temperature Cycling). All devices shall be subjected to thermal shock (temperature cycling) in accordance with MIL-STD-750, Method 1051, [illegible] at the temperature extremes shall be 15 minutes, minimum.

4.6.4 Acceleration. All devices shall be subjected to acceleration test in accordance with MIL-STD-750, Method 2006, with the following exceptions: The test shall be performed one time in the Y_1 orientation only, at a peak level of 20,000 G, minimum. The one minute hold-time requirement shall not apply.

4.6.5 Hermetic Seal Tests. All devices shall be subjected to hermetic seal tests (fine leak followed by gross leak) with test conditions as specified in 4.6.5.1 and 4.6.5.2. Failed devices from either test shall be removed from the lot.

4.6.5.1 Hermetic Seal (Fine-Leak) Test. All devices shall be fine-leaked tested in accordance with MIL-STD-750, Method 1071, test condition G or H. The leak-rate rejection criterion shall be 1×10^{-7} cubic centimeters of helium per second when measured at a differential pressure of one atmosphere.

4.6.5.2 Hermetic Seal (Gross-Leak) Test. All devices shall be tested for gross-leaks in accordance with MIL-STD-750, Method 1071, test condition A, B, C, D, or F.

4.6.6 Reverse Bias. All devices shall be subjected to reverse bias with the following test conditions and end-point measurement:

(a) The collector to base junction shall be reverse-biased at V_{CB} = 20 Vdc, I_E = 0, for 48 hours minimum at T_A = +150°C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:44Z
Check the source to verify that this is the current version before use.

4.6.6 Reverse Bias Continued.

(b) At the end of the high-temperature test time, the ambient temperature shall be lowered until T_C = +30 $\pm$5°C is attained.

(c) Remove the bias voltage and maintain all devices at room ambient temperature.

(d) Within 24 hours following bias removal, measure I_{CBO} as specified in Table V.

4.6.7 Pre Burn-In Tests. The parameters I_{CBO} and h_{FE} of Table V shall be measured on all devices in the lot. All devices shall be handled or identified such that the delta end-points can be determined after the burn-in test. All devices which fail to meet these requirements shall be removed from the lot and the quantity removed shall be noted on the lot history.

TABLE V. Burn-In Test Measurements

Test	Method	MIL-STD-750 Details	Symbol	Limits Min	Limits Max	Unit
Collector to Base cutoff current	3036	Bias Cond. D, V_{CB} = 15 Vdc	I_{CBO}	---	50	nAdc
Forward-current transfer ratio 2N6603 2N6604	3076	V_{CE} = 10 Vdc, I_C = 15 mAdc, I_C = 30 mAdc pulsed (see 4.4.1)	h_{FE}	30	200	----

4.6.8 Burn-In Test. All devices shall be operated for 168 hours minimum under the following conditions:

2N6603	T_A = 25 $\pm$3°C	$V_{CB} \geq$ 15 Vdc	P_T = 400 mW
2N6604	T_A = 25 $\pm$3°C	$V_{CB} \geq$ 15 Vdc	P_T = 500 mW

4.6.9 Post Burn-In Tests. The parameters I_{CBO} and h_{FE} of Table V shall be retested after burn-in on all devices in the lot. The parameters measured shall not have changed during the burn-in test from the initial value by more than the specified amount as follows:

Δh_{FE} = $\pm$ 15 percent

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:44Z
Check the source to verify that this is the current version before use.

TABLE I. Group A Inspection

Examination or Test	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Details	Non-TXV	TXV		Min	Max	
Subgroup 1			10	10				
Visual and mechanical examination	2071				--	--	--	--
Subgroup 2			5	5				
Breakdown voltage, collector to emitter	3011	Bias Cond. D I_C = 1.0mAdc, pulsed (see 4.4.1)			BV_{CEO}	15	--	Vdc
Breakdown voltage collector to base	3001	Bias Cond. D I_C = 100μAdc			BV_{CBO}	25		Vdc
Breakdown voltage, emitter to base	3026	Bias Cond. D I_E =100μAdc			BV_{EBO}	2	--	Vdc
Collector to base cutoff current	3036	Bias Cond. D, V_{CE} = 15Vdc			I_{CBO}	--	50	nAdc
Forward-current transfer ratio	3076	V_{CE} = 10Vdc			h_{FE}	30	200	--
2N6603		I_C = 15mAdc						
2N6604		I_C = 30mAdc pulsed (see 4.4.1)						
Subgroup 3			7	7				
Power-gain	---	V_{CE} = 10Vdc f = 1GHZ (see 4.4.2 and figure 5)			G_{pe}	14	20	db
2N6603		I_C = 15mAdc						
2N6604		I_C = 30mAdc						

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:44Z
Check the source to verify that this is the current version before use.

TABLE I. Group A Inspection

Examination or Test	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Details	Non-TXV	TXV		Min	Max	
Subgroup 3 Cont'd								
Collector-base capacitance	3236	V_{CB} = 10Vdc I_E = 0 .1MHz ≤f≤ 1MHz			C_{cb}			
2N6603						.25	.75	pf
2N6604						.45	.95	pf
Noise figure	---	V_{CE} = 10Vdc I_C = 5mAdc f = 1GHz (see 4.4.2 and figure 6)			NF			
2N6603						1.0	2.5	db
2N6604						1.5	3.0	db
Subgroup 4			10	10				
High-temperature operation:		T_A = +150°C						
Collector to base cutoff current	3036	Bias Cond. D, V_{CB} = 15Vdc			I_{CBO}	--	5.0	μAdc
Low-temperature operation:		T_A = -65°C						
Forward-current transfer ratio	3076	V_{CE} = 10Vdc			h_{FE}	15	--	--
2N6603		I_C = 15mAdc,						
2N6604		I_C = 30mAdc pulsed (see 4.4.1)						

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:44Z
Check the source to verify that this is the current version before use.

TABLE II. Group B Inspection

Examination or Test	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Details	Non-TXV	TXV		Min	Max	
Subgroup 1			20	20				
Physical dimensions	2066	(see Figure 1)			---	--	--	--
Subgroup 2			15	15				
Solderability	2026				---	--	--	--
Thermal shock (Temp. cycling)	1051	Test Cond. C 10 cycles; time @ temp. extremes = 15 minutes (min.) (see 4.2.3)			---	--	--	--
Thermal shock (glass strain)	1056	Test Cond. A			---	--	--	--
Seal (leak rate)	1071	Test Cond. G or H for fine leaks; test cond. A,B,C,D or F for gross leaks			---	--	$1x10^{-7}$	atm cc/sec
Moisture resistance	1021	Omit initial conditioning			---	--	--	--
End-points:								
Collector to base cutoff current	3036	Bias Cond. D, V_{CB} = 15Vdc			I_{CBO}	--	50	nAdc
Power-gain	---	V_{CE} = 10Vdc f = 1GHz (see 4.4.2 and Figure 5)			G_{pe}	14	20	db
2N6603		I_C = 15mAdc						
2N6604		I_C = 30mAdc						

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:44Z
Check the source to verify that this is the current version before use.

TABLE II. Group B Inspection

Examination or Test	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Details	Non-TXV	TXV		Min	Max	
Subgroup 3			15	15				
Shock	2016	Nonoperating; 1,500 G; 0.5 msec; 5 blows in each orientation: X_1, Y_1, Y_2, and Z_1			---	--	--	--
Vibration, variable frequency	2056				---	--	--	--
Constant acceleration	2006	20,000 G, in each orientation: X_1, Y_1, Y_2, and Z_1			---	--	--	--
End Points:								
(Same as Subgroup 2)								
Subgroup 4			20	20				
Terminal strength (lead fatigue) (electrical rejects may be used)	2036	Test Cond. A; Weight = 5 lbs Application time = 15sec each terminal			---	--	--	--
Seal (leak rate)	1071	Test Cond. G or H for fine leaks. Test Cond. A, B, C, D, or F for gross leaks			---	--	$1X10^{-7}$	atm cc/sec

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:44Z
Check the source to verify that this is the current version before use.

TABLE II. Group B Inspection

Examination or Test	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Details	Non-TXV	TXV		Min	Max	
Subgroup 5			7	7				
High-temperature life (non-operating)	1032	Tstg = 200°C (see 4.3.4)			---	--	--	--
End points:								
Collector-to base cutoff current	3036	Bias Cond. D, V_{CB} = -15Vdc			I_{CBO}	--	100	nAdc
Power-gain	---	V_{CE} = 10Vdc f = 1GHz (see 4.4.2 and Figure 5)			G_{pe}	14	20	db
2N6603		I_C = 15mAdc						
2N6604		I_C = 30mAdc						
Subgroup 6			7	7				
Steady-state operation life	1027	T_A = 25°C V_{CB} = ≥ 15Vdc			---	--	--	--
2N6603		P_T = 400mW						
2N6604		P_T = 500mW (see 4.3.4)						
End points;								
(Same as Subgroup 5)								

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:44Z
Check the source to verify that this is the current version before use.

TABLE III. Group C Inspection

Examination or Test	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Details	Non-TXV	TXV		Min	Max	
Subgroup 1			15	15				
Resistance to solvents (electrical rejects may be used)	--	Method 215 of MIL-STD-202 (see 4.4.3)			---	--	--	--
Subgroup 2			15	15				
Thermal shock	1051	Test Cond. C-1 time at temperature extremes = 15 minutes (min.) (see 4.2.3)			---	--	--	--
End points:								
(Same as Subgroup 2 of Group B)								
Subgroup 3			15	15				
Salt atmosphere (electrical rejects may be used)	1041							
Subgroup 4			λ=7	λ=7				
High-temperature life (non-operating)	1031	Tstg = +200°C (see 4.3.4)			---	--	--	--
End points:								
(Same as Subgroup 5 of Group B inspection)								

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:44Z
Check the source to verify that this is the current version before use.

TABLE III. Group C Inspection

Examination or Test	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Details	Non-TXV	TXV		Min	Max	
Subgroup 5			λ=7	λ=7				
Steady-state operation life	1026	T_A = 25°C V_{CB} = ≥ 15Vdc						
2N6603		P_T = 400mW						
2N6604		P_T = 500mW						
End points:								
(Same as Subgroup 5 inspection)								
Subgroup 6			15	15				
Power-gain at optimum noise figure	---	V_{CE} = 10Vdc f = 1GHz (See 4.4.2 and Fig. 5&6)			G_{NF}			
2N6603		I_C = 5mAdc				10	--	db
2N6604		I_C = 5mAdc				9	--	db
Subgroup 7			15	15				
Input reflection coefficient	---	V_{CE} = 10Vdc f = 1GHz Z_O = 50Ω						
2N6603		I_C = 15mAdc			/S11/	.5	.8	--
					LØS11	+165	-165	deg.
2N6604		I_C = 30mAdc			/S11/	.6	.9	--
					LØS11	+160	-170	deg.
Output reflection Coefficient	---	V_{CE} = 10Vdc f = 1GHz Z_O = 50Ω						
2N6603		I_C = 15mAdc			/S22/	.3	.5	--
					LØS22	-55	-25	deg.
2N6604		I_C = 30mAdc			/S22/	.1	.3	--
					LØS22	-95	-65	deg.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:44Z
Check the source to verify that this is the current version before use.

TABLE III. Group C Inspection

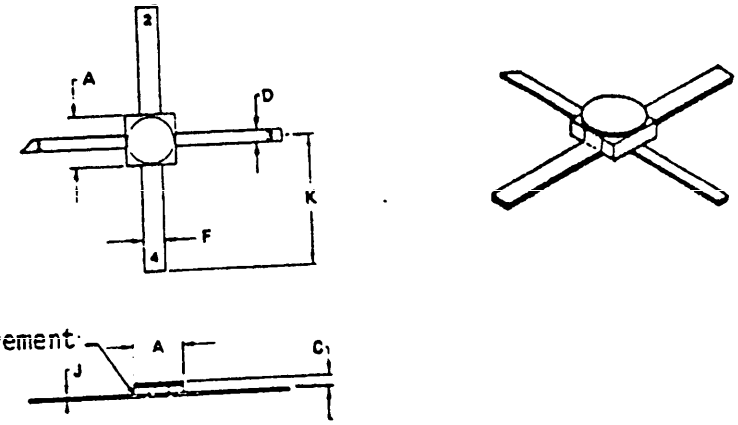
Examination or Test	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Details	Non-TXV	TXV		Min	Max	
Subgroup 7 Cont'd								
Forward transfer coefficient	---	V_{CE} = 10Vdc f = 1GHz Z_O = 50Ω						
2N6603		I_C = 15mAdc			$\|S21\|^2$	12	--	db
2N6604		I_C = 30mAdc			$\|S21\|^2$	12	--	db
Reverse transfer coefficient		V_{CE} = 10Vdc f = 1GHz Z_O = 50Ω						
2N6603		I_C = 15mAdc			$\|S12\|^2$	--	-22.5	db
2N6604		I_C = 30mAdc			$\|S12\|^2$	--	-22.5	db
Subgroup 8			10	10				
Noise figure	---	V_{CE} = 10Vdc I_C = 5mAdc f = 2GHz (see 4.4.2 and Figure 6)			NF			
2N6603						--	4	db
2N6604						--	4.5	db
Power gain at optimum noise figure	---	V_{CE} = 10Vdc I_C = 5mAdc f = 2GHz (see 4.4.2 and Figures 5 & 6)			G_{NF}	6	--	db

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:44Z
Check the source to verify that this is the current version before use.

TABLE III. Group C Inspection

Examination or Test	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Details	Non-TXV	TXV		Min	Max	
Subgroup 8 Cont'd								
Power gain	---	V_{CE} = 10Vdc f = 2GHz (see 4.4.2 and Figure 5)			G_{pe}			
2N6603		I_C = 15mAdc				7	--	db
2N6604		I_C = 30mAdc				8	--	db

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:44Z
Check the source to verify that this is the current version before use.



NOTE: The top of the device body is hermetically sealed by a metallic cap (shaded region) which is electrically connected to the emitter.

DIM	MILLIMETERS		INCHES	
	MIN	MAX	MIN	MAX
A	2.29	2.67	0.090	0.105
C	0.89	1.40	0.032	0.050
D	0.41	0.61	0.016	0.024
F	0.89	1.09	0.035	0.043
J	0.08	0.15	0.003	0.006
K	4.45	5.84	0.175	0.230

PIN 1. COLLECTOR
2. EMITTER
3. BASE
4. EMITTER

FIGURE 1. Physical dimensions of Transistor types 2N6603 and 2N6604.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T02:44Z
Check the source to verify that this is the currentversion before use.

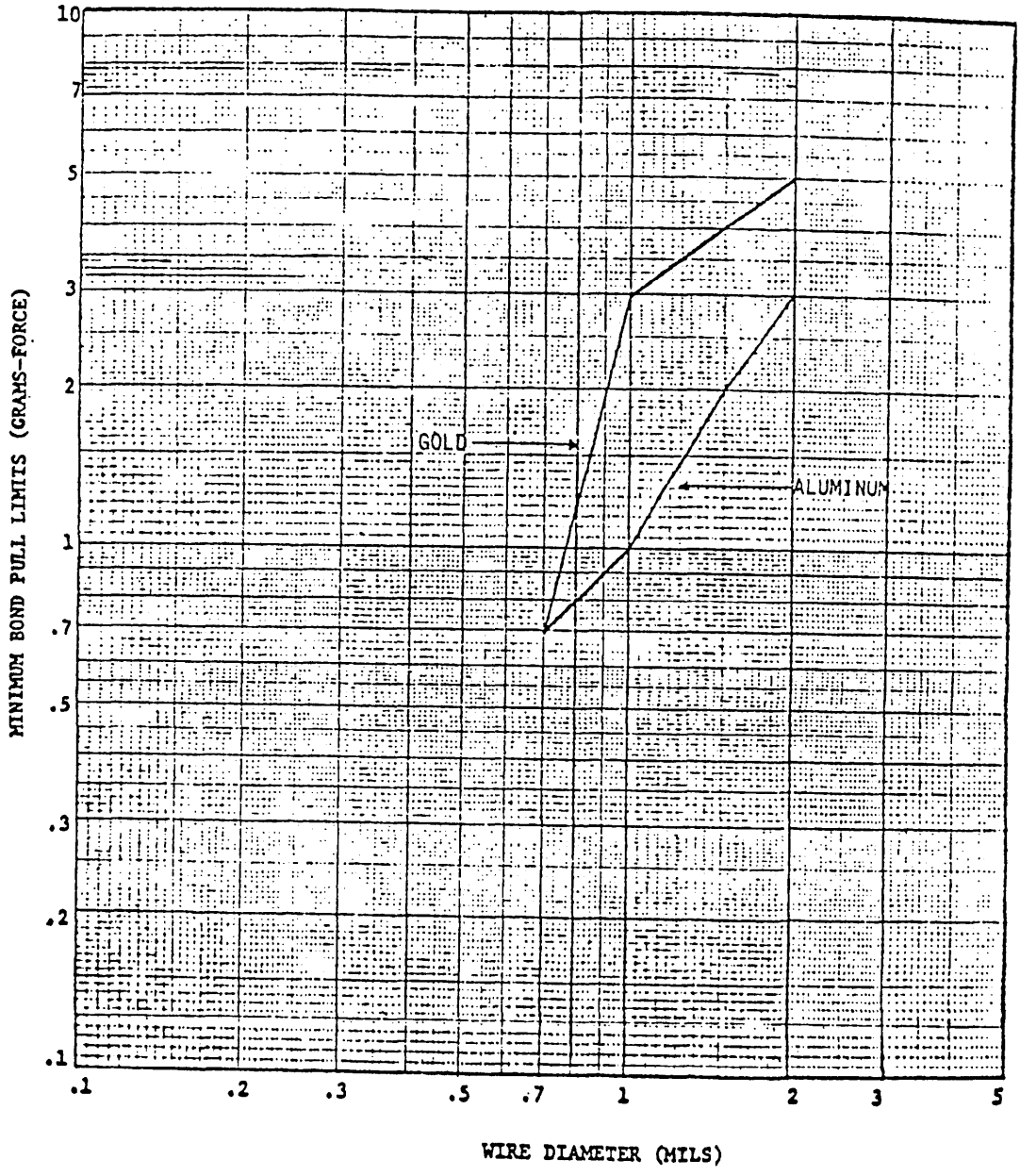
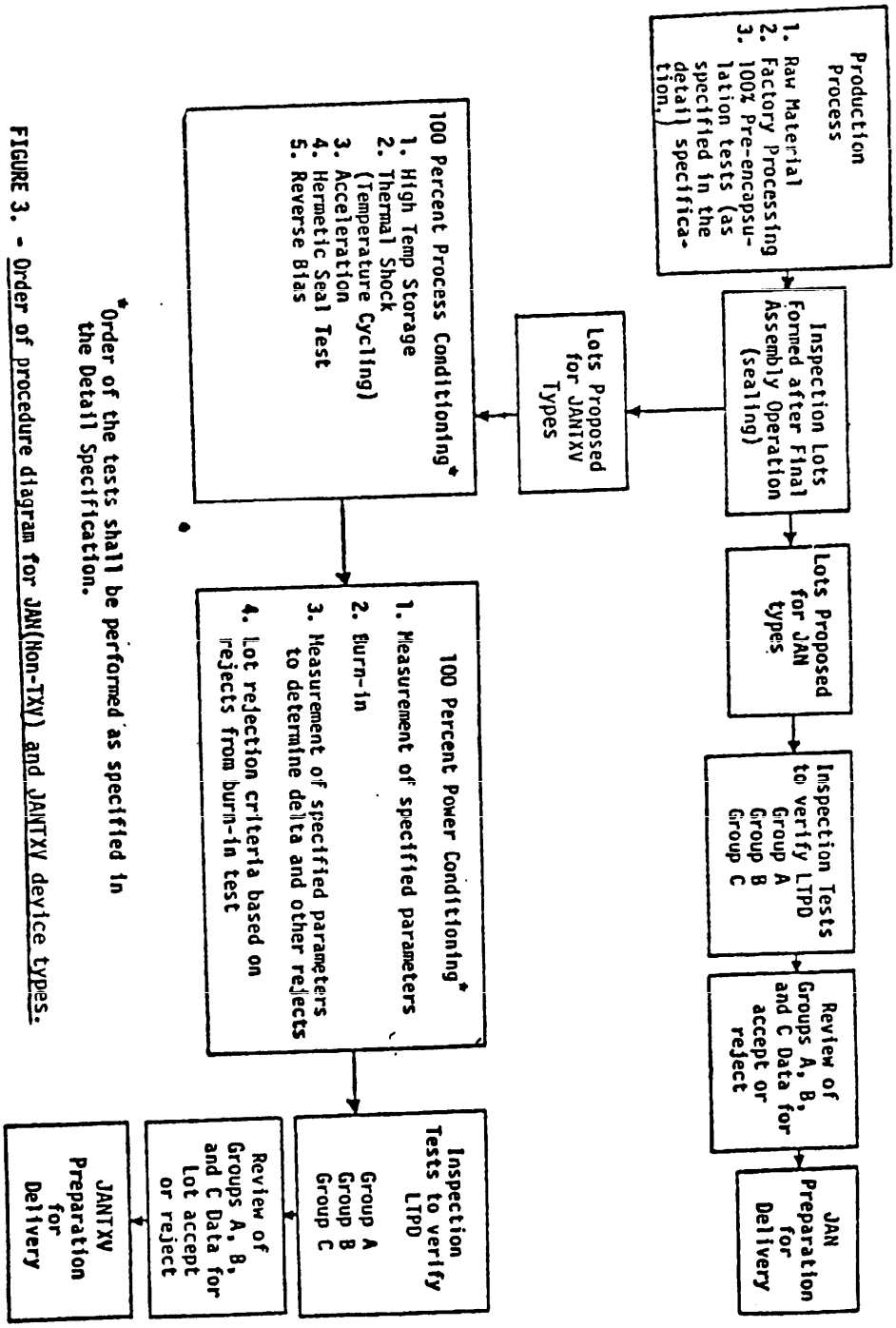


Figure 2. Minimum destructive bond-pull breaking-force vs. wire diameter for wire bonds.

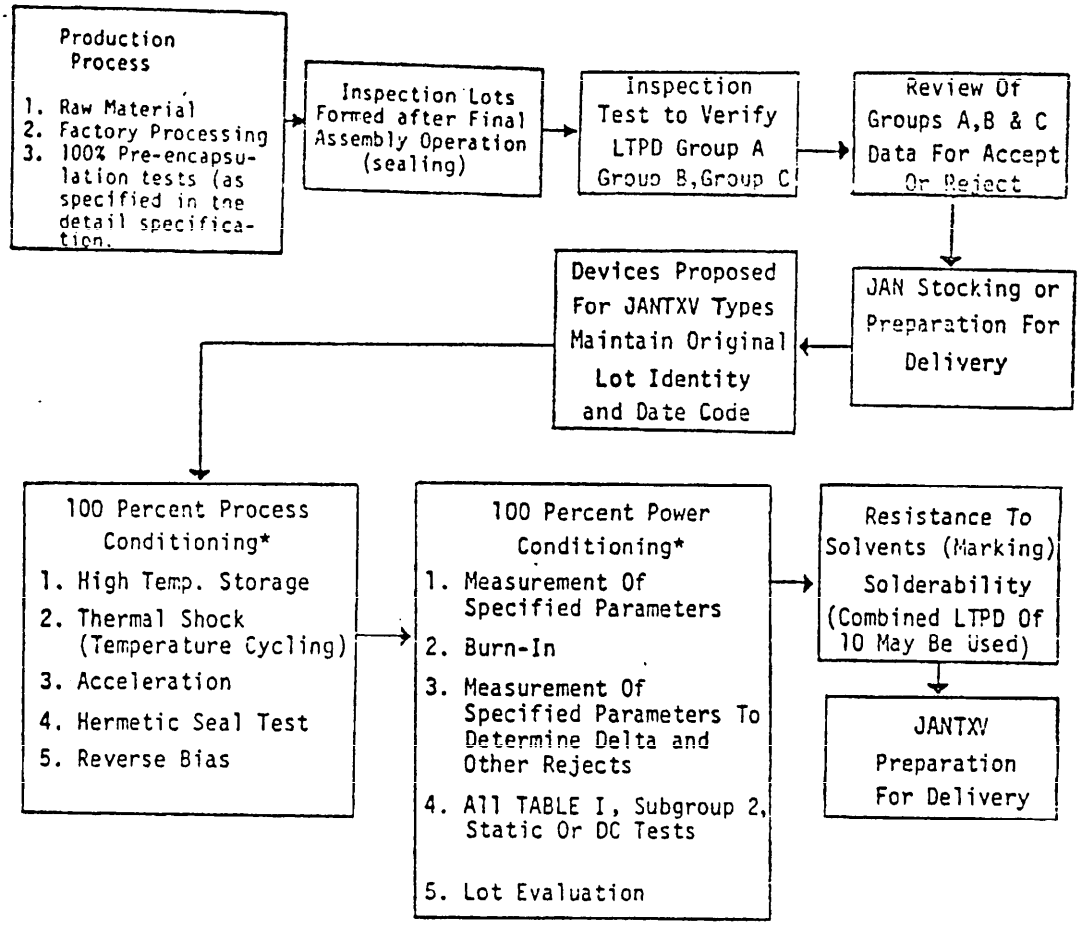
Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:44Z
Check the source to verify that this is the current version before use.



*Order of the tests shall be performed as specified in the Detail Specification.

FIGURE 3. - Order of procedure diagram for JAN(Non-TXV) and JANTXV device types.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:44Z
Check the source to verify that this is the current version before use.



*ORDER OF THE TESTS IN THE BLOCKS SHALL BE PERFORMED AS SHOWN

FIGURE 4. Alternate order of procedure diagram for JAN (Non-TXV) and JANTXV device types.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:44Z
Check the source to verify that this is the current version before use.

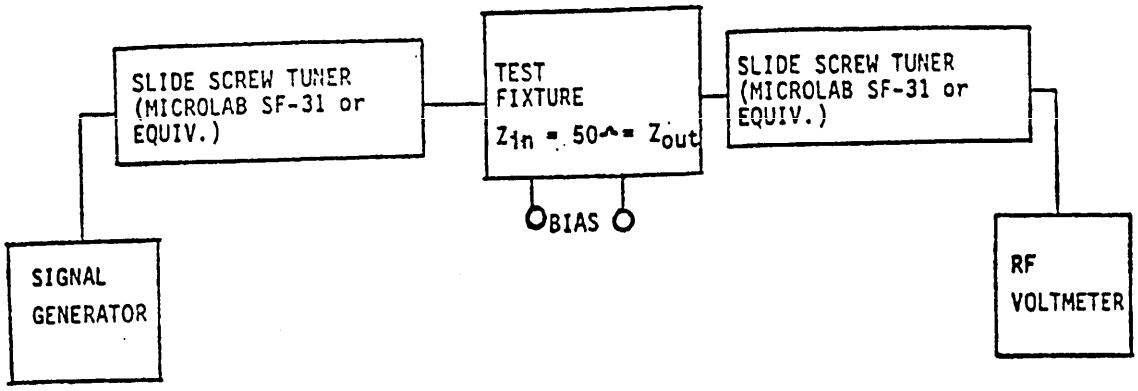


Figure 5: Block Diagram of Power Gain Test Circuit at 1 GHz.

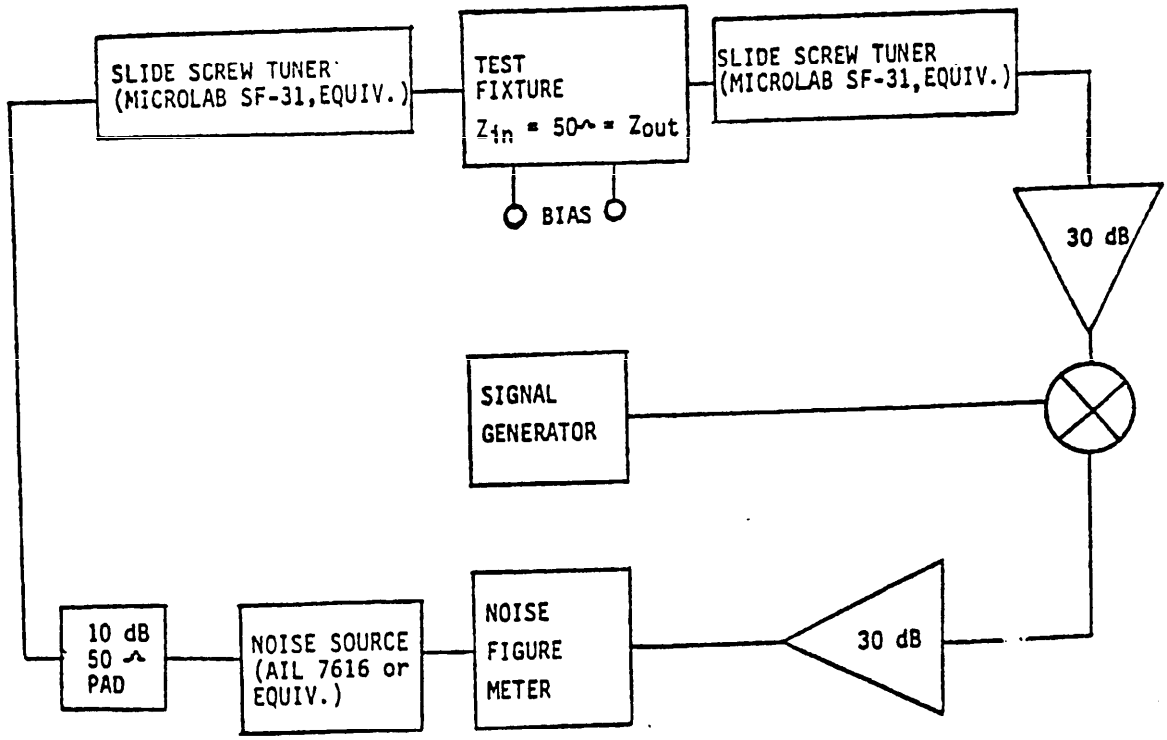


Figure 6: Block Diagram of Noise Figure Test Circuit at 1 GHz.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:44Z
Check the source to verify that this is the current version before use.

4.6.9.1 Burn-In Test Failures (Screening). All devices that exceed the delta (Δ) limits of 4.6.9 or the limits of Table V after burn-in, shall be removed from the inspection lot and the quantity removed shall be noted on the lot history. If the quantity removed after burn-in should exceed 10 percent of the total inspection lot on the burn-in test, then the entire lot shall be unacceptable as TXV types. However, if the quantity removed is between 10 and 20 percent of the total inspection lot on the burn-in test, the lot may be resubmitted for burn-in one time only and shall contain only parts which were in the original lot. Resubmitted lots shall be kept separate from new lots and shall be inspected for all the specified characteristics of 4.6.9 and Table V using a tightened inspection PDA of 5 percent. If the PDA of 5 percent is exceeded for the resubmitted lot, the entire lot shall be unacceptable for TXV types.

4.6.9.2 Electrical Inspection. If the option provided in Figure 4 is used, all devices in the lot must also be submitted to all static or DC tests specified in Table I, Subgroup 2. All units failing these tests must be removed from the lot and the quantity removed shall be noted on the lot history. If the quantity removed exceeds 7 percent, the lot shall be unacceptable for "TXV" types.

4.7 Resistance to Solvents and Solderability Tests. If the option provided in Figure 4 is taken, a sample of devices from the lot must be submitted to the tests of Group C, Subgroup 1 (resistance to solvent) and Group B, Subgroup 2 (solderability only) following 100 percent processing. A sample associated with the LTPD's of the appropriate subgroup or a combined sample for both subgroups using an LTPD of 10 shall be used. These tests are to be conducted one time only on a given lot and shall be conducted on the first sublot of 100 percent processed product. Electrical rejects from the post burn-in inspection may be used for these tests.

5. PREPARATION FOR DELIVERY

5.1 See MIL-S-19500, Section 5.

Custodian:
Army - EL

Preparing Activity:
Army - EL

(Project 5961-A675)

Source: https://assist.dla.mil -- Downloaded: 202610-19T02:44Z
Check the source to verify that this is the current version before use.

To detach this form, cut along this line

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-S-19500/522(EL) SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, HIGH-FREQUENCY DEVICE TYPES: JAN 2N6603 AND TXV 2N6603

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:44Z
Check the source to verify that this is the current version before use.

FOLD

Department of the Army
Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

POSTAGE AND FEES PAID

OFFICIAL BUSINESS

Commanding General
U.S.Army Electronics Command
ATTN: ~~AMSEL~~ DRSEL-RD-TS
Fort Monmouth, New Jersey 07703

To detach this [illegible], cut along this line

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026l0-19T02:44Z
Check the source to verify that this is the current version before use.